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Successful Application of Advanced Enzyme Clean Up System in Perforated Completions to Revive Dead Wells – Case Studies from Romania

D. Ghosh, B. Ghosh, A. R. Mohammed, H. Moneb, P. Bansal, and D. J. V. Trono, Epygen Labs FZ LLC, Dubai, U.A.E; A. Dragomir and A. Matei, OMV Petrom, Romania

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Abstract

The presence of residual mud cake around perforation tunnels is a key factor contributing to production decline and blockage of reservoir fluid flow into the wellbore, even after conventional acid stimulation procedures. This work extends previous work on conventional enzymatic cleanup, highlighting the capabilities of a newly developed recombinant enzyme that has proven efficient in perforated completions, where damage is caused by drilling and perforating processes.

Specially developed cellulase enzymes break down cellulose-based biopolymers into simpler sugars, improving fluid flow. This enzyme, combined with bio-acetic acid and bio-surfactant, is injected at perforated points into the formation to target clogging materials such as biopolymers and carbonates. In laboratory trials, enzymatic percolation through porous media was studied using mud cake built on ceramic discs using an HPHT setup, along with reducing sugar assay estimates. Field trials conducted in a Romanian oil field involved using the treatment solution in wells drilled with cellulose-based mud. A 12-hour seepage through the tunnel pores allowed the dissolution of the internal cakes. The tailored enzyme system effectively cleaned the wellbore, offering a sustainable alternative for damage clean-up.

The wells were first perforated in 2014, but oil production declined abruptly after achieving the maximum flow potential, eventually becoming completely non-productive. Despite two periodic hydrochloric acid stimulation treatments on both wells, no long-term benefits were achieved. While partial production enhancement was observed, it remained at a rate lower than the wells' optimum flow potential. Subsequently, a rapid reduction in flow led to well closure within a period of three to twelve months. Post-treatment steady-state flow rates showed a 40% improvement in flow profiles compared to conventional treatments conducted on analogous wells. The results proved substantial clean-up, reviving the wells and maintaining production for over 12 months to date.

This novel application of the specialized cellulase enzyme system establishes an effective perforation cleanup method for the dissolution of compressed mud cake in perforated zones. This system could be a primary choice for removing damage in perforation tunnels and internal filter cakes, ultimately improving well injectivity and productivity.

Introduction

Dirty perforations around the perforation tunnel often impact a well's inflow performance. The deposition of filter cake in the perforation tunnels and the presence of internal filter cake can cause pore plugging, which may partially impair well performance. The purpose of this paper is to extend the earlier works on enzyme applications, highlighting the enhanced capabilities of an enzymatic cleanup solution in perforated completions where one of the permeability damage mechanisms could be the compaction of residual mud cake polymers, which could not be removed by the action of previously applied HCl solution.

Enzymes are biocatalysts that accelerate chemical reactions without being consumed. An enzyme's function is intrinsically linked to its three-dimensional structure, which defines how it achieves substrate binding, catalysis, and regulation. Enzymes are commonly used as mud-cake breakers in open-hole limestone and carbonate formations because they target the bio-polymer component of damaging mud cakes, which result from the systematic degradation of drilling mud cake. An enzyme with controlled reaction activity can be tailored to specific well conditions and drilling fluid properties, making it a preferred wellbore cleanup system.

This unique cleanup process applied in two dead wells in Romanian wells exemplifies the efficiency of the advanced enzyme system in the specific dissolution of mud cake across the perforated interval, successfully bringing the dead wells back into production.

Methodology

Materials and Methods

- HT-HP Fluid loss apparatus having an arrangement to hold 2.5" diameter ceramic discs
- Heating Jacket for HT- HP fluid loss cell
- Nitrogen cylinder with manifold
- Ceramic discs (mean pore dia=10 microns)
- Ceramic discs (mean pore dia=50 microns)
- Filtered 2.0% KCl brine
- Drilling mud sample- collected from a drill-site
- Treating solution containing test enzyme.
- Precision weighing balance
- Lugol's reagents
- 3,5-Dinitrosalicylic acid (DNS)
- Potassium sodium tartrate tetrahydrate

HPHT Apparatus

The HPHT fluid loss test is conducted to generate the filter cake of a drilling fluid under downhole conditions. It features a heating jacket that allows the specific drilling fluid sample to be heated to the wellbore temperature, while the desired differential pressure is applied to create the filter cake on ceramic discs for enzyme percolation studies.

Filter-Cake Formation in HPHT Set Up for Enzyme Percolation Studies

- Ceramic discs were labeled, and the weight of each disc was recorded.
- The discs were immersed in a 2% KCl solution (20 g KCl in 1 L of DM water) for 2 hours and mounted into the fluid loss cell, keeping the bottom valve closed.
- A 20 ml sample of drill-in fluid was poured directly into the cell.
- The cell temperature was set to 55°C and allowed to stabilize.
- The top valve of the cell was opened, and 100 psi pressure was applied using a nitrogen cylinder.
- The filter cake was prepared by running the fluid loss test for 30 minutes. Afterwards the pressure was released, and the remaining mud was slowly decanted and washed with DM water to remove residual fluid.
- The disc with the filter cake was carefully removed along with the bottom assembly to ensure the filter cake structure remained intact.
- The filter cake formed was used for the enzyme percolation studies described in Section 3.4.

Engineered Cellulase Enzyme Percolation through Filter Cake

- The enzyme percolation experiments were conducted inside a hot-air oven at 55°C.
- Air was used as the non-wetting phase, while the wetting liquid consisted of the enzyme-based clean-up fluid (CUF). This formulation included:
 - 6 wt.% CL Enzyme
 - 1 wt.% Bio-surfactant, all mixed into 2 wt.% KCl brine
- The CUF solution (250 ml) was placed beneath a ceramic disc, which had a thin layer of filter cake on its surface and significant internal damage.
- The disc was half-immersed in the CUF solution and kept inside the hot-air oven for 8 hours at 55°C (Fig. 1), allowing the CUF to percolate through the disc.
- During the soaking period, cake samples were harvested at multiple time intervals: 0 hrs, 2 hrs, 6 hrs, and 8 hrs by scraping the disc surface, allowing the remaining cake to continue soaking. Reducing sugar estimation was performed on the scraped samples using the standard reducing sugar method (Miller, G.L., 1959) with DNS reagent and UV spectrophotometry. This process evaluates the percolation action of the CUF, which hydrolyzes the cellulose-based biopolymer components of the filter cake into simple sugars.
- The results were plotted for reducing sugar release against time intervals (Fig. 3).

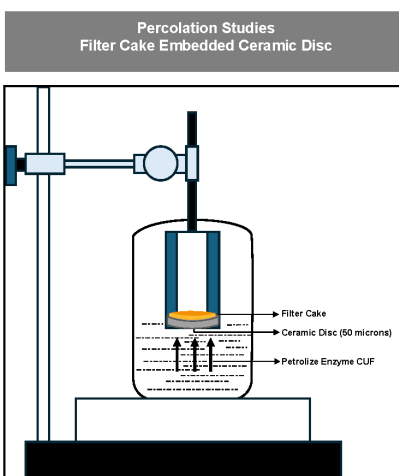


Figure 1—Schematic diagram – CUF Percolation Studies (adapted from Alyafei, N. & Blunt, M.J., 2018)

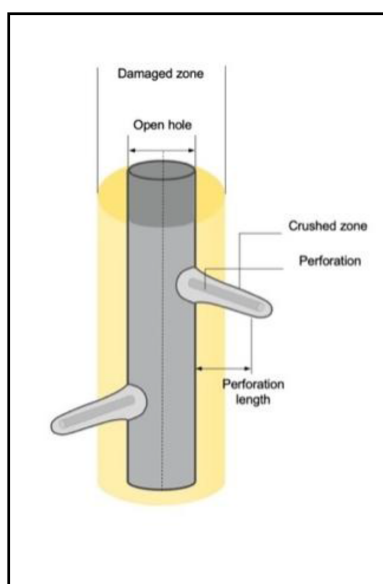


Figure 2—Schematic diagram – of Perforation Damage (Asuelimen and Adetona, 2013)

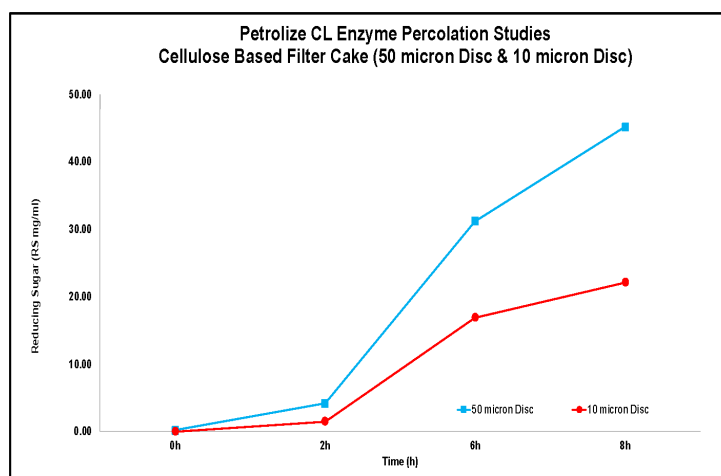


Figure 3—Comparative evaluation of reducing sugar on a cellulose based filter cake embedded ceramic disc (10 micron & 50 micron)

Evaluation of reducing sugar released from cellulose-based filter-cake by CUF action.

In the event of well perforation, additional permeability reduction may occur, which can be classified as follows (Fig. 2):

- **Crushed Zone:** Mechanical deformation and compaction of the rock around the perforation tunnel.
- **Debris:** Debris from the perforation charge and formation can clog pore spaces.
- **Mud Cake Invasion:** Drilling mud cake and filtrate invade the formation, reducing permeability.

Laboratory experiments demonstrate that the enzyme system would percolate through the perforation-damaged zone and the enzyme protein retains its bioactivity. It is worth noting that the configuration of the enzyme protein can be affected by various forces, including, but not limited to, heat energy, which influences the structural bonds and 3D configuration of the protein. In this experiment, the enzyme solution is allowed to percolate through the disc without any driving pressure difference, to observe the effect of the percolated enzyme on the mud ingredients, a representation of the perforation-damaged zone. Though the percolation rate would differ significantly in the perforated zones because of applied fluid pressure and differences in permeability, this simple procedure can be used to measure the effect of enzymatic fluid on the pore damage. When the enzyme percolates through the pores and comes in contact with the cellulose component of the filter cake on the other side of the disc, it breaks down the long-chain cellulose polymer (PAC) into simple sugar. It is expected that more cellulase enzyme will percolate through the perforation-damaged zone due to added pressure and depolymerize long-chain compacted cellulose polymers. Also, the presence of mild acid in the system would help to dissolve salt (mainly calcium) deposition, if any, resulting in a better perforation cleanup. The reducing sugar readings are tabulated and plotted with respect to time.

Reducing Sugar Estimation using DNS Method

The DNS assay test is used to measure the release of sugars from a cellulosic substrate (such as cellulose-based filter cake) by the action of the enzyme.

Preparation of 3, 5 – Dinitrosalicylic Acid Solution (DNS)

- Suspend 20.0 grams of 3,5-Dinitrosalicylic acid in 1000 mL of distilled water and gradually add 300 mL of a 10.67% (w/v) sodium hydroxide solution while mixing.
- Warm the suspension in a water bath at 50°C until the solution becomes clear.
- Gradually add 600 grams of potassium sodium tartrate tetrahydrate to the solution with continuous mixing.
- Dilute the solution to 2000 mL with distilled water and filter through a coarse sintered glass filter.
- Store the solution in a dark amber bottle at room temperature.

Reducing Sugar Estimation Procedure

- 0.5 g of cellulose-based filter-cake samples were taken at time intervals of 0 h, 2 h, 6 h, and 8 h in autoclave-grade Falcon tubes containing 9.5 mL of sodium acetate buffer (pH 4.8).
- The release of reducing sugars was estimated by the DNS method as follows:
 - The Falcon tubes containing filter-cake samples were centrifuged at 3000 rpm for 20 minutes.
 - 0.8 mL of the supernatant was taken for reducing sugar estimation by the DNS method.

- 1.2 mL of DNS solution was added to 0.8 mL of the supernatant sample, capped, and vortexed. The solution was kept at room temperature for 30 minutes, followed by DNS reaction carried out in a water bath at 95°C for 15 minutes.
- After the DNS reaction, the samples were kept in a water bath for 10 minutes at room temperature. Afterward, they were kept at room temperature for 10 minutes. The absorbance of the control and test samples was measured at 540 nm against distilled water as blank and DNS reagent blank using a Shimadzu spectrophotometer.
- Calculation - Reducing sugar (Glucose Equivalent).

$$\frac{1}{a} \times (\text{Final : dilution}) \times \left[\frac{\Delta \text{Abs} + (\Delta \text{Abs} + Y \text{Intercept})}{\frac{1}{a}} \right] \quad (\text{Eqn 1})$$

Field Application

Location: Romania

Well details.

Well #1

- Vertical well
- Cased hole perforated well completion
- Top Perforations = 1003 m
- Bottom of perforations = 1046 m
- Effective perforations – 22 m (6 intervals)

Well #2

- Vertical well
- Cased hole perforated well completion
- Top Perforations = 1062 m
- Bottom of perforations = 1088 m
- Effective perforations – 26 m (5 intervals)

Drilling Mud

- Biopolymer (xanthan gum) = 3 kg/m³
- Polymer to control fluid loss and rheology (polyanionic cellulose) = 1– 2 kg/m³
- Polymer to control fluid loss (ultra-low polyanionic cellulose) = 7 – 6 kg/m³
- Calcium carbonate = 100 – 150 kg/m³
- Low gravity solids = 6.4 kg/m³
- KCl = 54000 ppm

Well Histories

In 2014-15, two cased hole vertical wells were in sandstone reservoirs and completed with perforations in six sand layers. Oil production declined within a short span after peaking to maximum flow until production fell below the economic level. Well #1 was tied up in 2015 with a peak production of 1.3 ton/d before falling within 3 months. Well #2 began production in 2014 at 4.7 ton/d, falling within 12 months to nearly zero. Operator conducted a conventional stimulation perforation clean up job with 15% HCl wells started to flow. Whereas the flow for Well#1 was nearly constant for about 15 months before ceasing, Well#2 produced at a constantly declining rate for about 1.5 years before ceasing flow. The reason for the production decline could be attributed to many factors, such as movement of formation fines, deposition of scale or for release, relocation and swelling of compacted biopolymers causing further damage. From close observations and analysis of production behavior the operator predicted that the presence of compacted mud cake around the perforated interval could be the primary reason for the blockage of reservoir fluid flow. It is well known that although HCl can very well dissolve the carbonates present in mud cake, it fails to dissolve the biopolymers. Thus, instead of a repeat HCl stimulation, the operator decided to go for the unconventional method of perforation cleanup using enzymatic fluid.

Technical Approach

Mixing treatment fluids

The Engineered Enzyme, along with other chemicals were mixed in the field as per ratios tested in the laboratory. The main-flush solution of 23 m³ of solution based on enzymes was prepared with the following composition as per Table 1. An Over-flush Solution of 5 m³ was prepared with the following recipe as per Table 2.

Table 1—Composition of cleanup fluid.

Chemical Product	Purpose of use	Conc. (%)	Quantity	Unit	
				L/kg	Barrels
Filtered Fresh water	Basic fluid		20690	L	
Ammonium Chloride	Clay stabilizer	5	1150	kg	
Engineered Cellulase	Cellulase Enzymes	6	1380	L	6.5
Bio-Acetic Acid	Organic Acid	3	700	L	
Bio-Surfactant	Bio-Surfactant	1	230	L	1.5

Table 2—Field preparation for Over-flush solution

Chemical Product	Purpose of use	Conc.(%)
Filtered Fresh water	Basic fluid	
Emulsifier	Demulsifier	0.5
Ammonium Chloride	Clay stabilizer	5

Operational program for field application of CUF

- 5-inches retrievable mechanical packer with 2 7/8 inches tubing installed and completed, followed by a pressure test.
- In the main flush, 23 m³ of CUF was pumped through the tubing.

- As over-flush solution, the CUF was displaced with tubing volume of 5m³ of water treated with ammonium chloride.
- Pumping pressure at the pumping unit did not exceed 36 bar, maintaining a pumping rate of 150 LPM (corresponding to the formation fracturing pressure).
- The well was Shut-in for 12 hours for percolation and breaking reaction.
- The spent CUF was circulated out from the well-bore with filtered brine.
- The well was put to production.

Results and Discussion

After the initial production tests, production data from Well #1 indicated a jump of 45 bar in stable wellhead pressure, whereas Well #2 showed only a small increase in wellhead pressure initially. However, the liquid flow rates for Well #2 were observed to have an adequate increase. During the 3-week observation period, Well #1 (Fig. 4) produced 8 m³/D initially and stabilized at 7 m³/D, indicating the revival of the dead well.

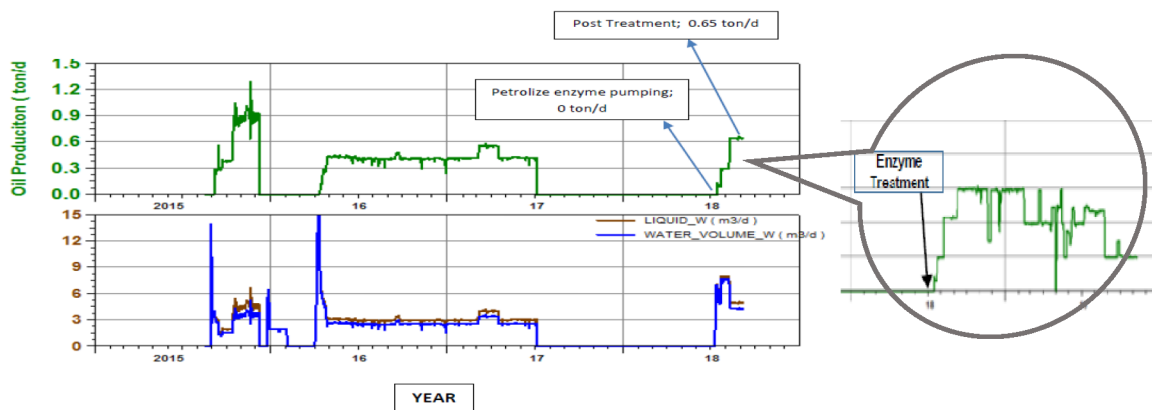


Figure 4—Well #1 Pre & Post treatment oil production data

Production data from Well #2 also show recovery of the dead well, producing about 8 m³/D of liquid with about 3 m³/D of oil, which stabilized at about 2.5 m³/D (Fig. 5).

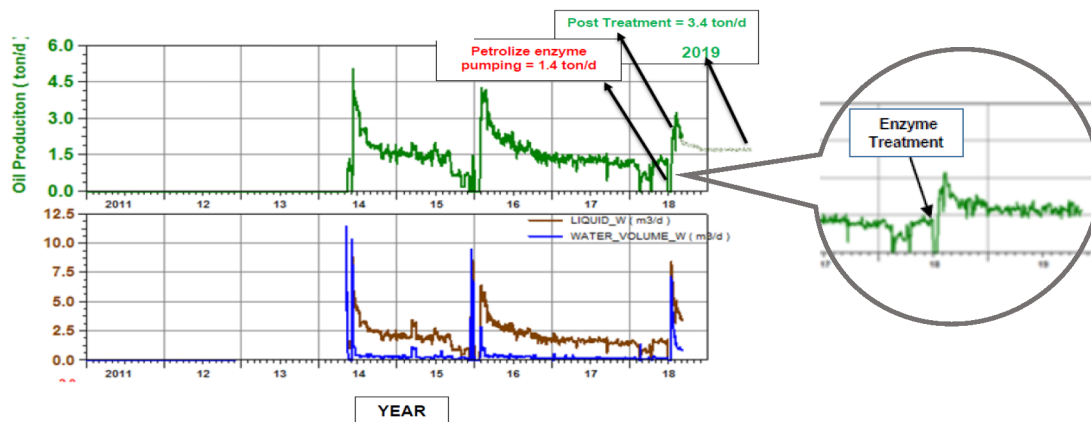


Figure 5—Well #2 Pre & Post treatment oil production data

Conclusions

Two cased-hole perforated wells were selected by the operator for application of engineered cellulase enzyme-based clean-up treatment on the entire perforation interval, with the objective of removing residual mud cake and other damages within the perforation tunnels. Post-enzyme treatment, the wells showed encouraging activity and steady flow.

For the dead well (Well #1), steady production is now observed, which is about 35% higher than the previously recorded flow rate. Well #2 showed even better results, with production more than double the previous average.

It is therefore concluded that this engineered enzyme system is a suitable solution for removing perforation tunnel damage if caused by biopolymer residues and can revive/improve productivity in a cost-effective manner.

Nomenclature

CL	Cellulase
LPM	Liter Per Minute
DP	Drill Pipe
PAC	Polyanionic Cellulose
HCL	Hydrochloric Acid
KCL	Potassium Chloride
DM	Demineralized Water

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